

•40 An air-filled parallel-plate capacitor has a capacitance of 1.3 pF. The separation of the plates is doubled, and wax is inserted between them. The new capacitance is 2.6 pF. Find the dielectric constant of the wax.

•42 A parallel-plate air-filled capacitor has a capacitance of 50 pF. (a) If each of its plates has an area of  $0.35 \text{ m}^2$ , what is the separation? (b) If the region between the plates is now filled with material having  $\kappa = 5.6$ , what is the capacitance?

•45 A certain parallel-plate capacitor is filled with a dielectric for which  $\kappa = 5.5$ . The area of each plate is  $0.034 \text{ m}^2$ , and the plates are separated by 2.0 mm. The capacitor will fail (short out and burn up) if the electric field between the plates exceeds 200 kN/C. What is the maximum energy that can be stored in the capacitor?

•32 A parallel-plate air-filled capacitor having area  $40 \text{ cm}^2$  and plate spacing 1.0 mm is charged to a potential difference of 600 V. Find (a) the capacitance, (b) the magnitude of the charge on each plate, (c) the stored energy, (d) the electric field between the plates, and (e) the energy density between the plates.

•8 How many  $1.00 \mu\text{F}$  capacitors must be connected in parallel to store a charge of 1.00 C with a potential of 110 V across the capacitors?

•9 Each of the uncharged capacitors in Fig. 25-27 has a capacitance of  $25.0 \mu\text{F}$ . A potential difference of  $V = 4200 \text{ V}$  is established when the switch is closed. How many coulombs of charge then pass through meter A?

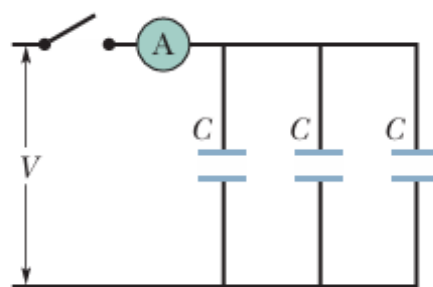


Figure 25-27 Problem 9.

## CURRENT ELECTRICITY

**24.38.** A parallel-plate capacitor has capacitance  $C_0 = 5.00 \text{ pF}$  when there is air between the plates. The separation between the plates is  $1.50 \text{ mm}$ . (a) What is the maximum magnitude of charge  $Q$  that can be placed on each plate if the electric field in the region between the plates is not to exceed  $3.00 \times 10^4 \text{ V/m}$ ? (b) A dielectric with  $K = 2.70$  is inserted between the plates of the capacitor, completely filling the volume between the plates. Now what is the maximum magnitude of charge on each plate if the electric field between the plates is not to exceed  $3.00 \times 10^4 \text{ V/m}$ ?

**25.47.** Consider a resistor with length  $L$ , uniform cross-sectional area  $A$ , and uniform resistivity  $\rho$  that is carrying a current with uniform current density  $J$ . Use Eq. (25.18) to find the electrical power dissipated per unit volume,  $p$ . Express your result in terms of (a)  $E$  and  $J$ ; (b)  $J$  and  $\rho$ ; (c)  $E$  and  $\rho$ .

**25.49.** Consider the circuit of Fig. 25.37. (a) What is the total rate at which electrical energy is dissipated in the  $5.00\text{-}\Omega$  and  $9.00\text{-}\Omega$  resistors? (b) What is the power output of the  $16.0\text{-V}$  battery? (c) At what rate is electrical energy being converted to other forms in the  $8.0\text{-V}$  battery? (d) Show that the power output of the  $16.0\text{-V}$

**25.42.** A resistor with a  $15.0\text{-V}$  potential difference across its ends develops thermal energy at a rate of  $327 \text{ W}$ . (a) What is its resistance? (b) What is the current in the resistor?

**25.43. Light Bulbs.** The power rating of a light bulb (such as a  $100\text{-W}$  bulb) is the power it dissipates when connected across a  $120\text{-V}$  potential difference. What is the resistance of (a) a  $100\text{-W}$  bulb and (b) a  $60\text{-W}$  bulb? (c) How much current does each bulb draw in normal use?

**25.49.** A battery-powered global positioning system (GPS) receiver operating on 9.0 V draws a current of 0.13 A. How much electrical energy does it consume during 1.5 h?

The resistance between its ends is  $0.104\ \Omega$ . (a) What is the resistivity of the material? (b) If the electric-field magnitude in the conductor is  $1.28\ \text{V/m}$ , what is the total current? (c) If the material has  $8.5 \times 10^{28}$  free electrons per cubic meter, find the average drift speed under the conditions of part (b).

**25.58.** A plastic tube 25.0 m long and 4.00 cm in diameter is dipped into a silver solution, depositing a layer of silver 0.100 mm thick uniformly over the outer surface of the tube. If this coated tube is then connected across a 12.0-V battery, what will be the current?

**25.68.** The potential difference across the terminals of a battery is 8.4 V when there is a current of 1.50 A in the battery from the negative to the positive terminal. When the current is 3.50 A in the reverse direction, the potential difference becomes 9.4 V. (a) What is the internal resistance of the battery? (b) What is the emf of the battery?



**25.10.** (a) At room temperature what is the strength of the electric field in a 12-gauge copper wire (diameter 2.05 mm) that is needed to cause a 2.75-A current to flow? (b) What field would be needed if the wire were made of silver instead?

**25.11.** A 1.50-m cylindrical rod of diameter 0.500 cm is connected to a power supply that maintains a constant potential difference of 15.0 V across its ends, while an ammeter measures the current through it. You observe that at room temperature (20.0°C) the ammeter reads 18.5 A, while at 92.0°C it reads 17.2 A. You can ignore any thermal expansion of the rod. Find (a) the resistivity and (b) the temperature coefficient of resistivity at 20°C for the material of the rod.

**25.12.** A copper wire has a square cross section 2.3 mm on a side. The wire is 4.0 m long and carries a current of 3.6 A. The density of free electrons is  $8.5 \times 10^{28}/\text{m}^3$ . Find the magnitudes of (a) the current density in the wire and (b) the electric field in the wire. (c) How much time is required for an electron to travel the length of the wire?